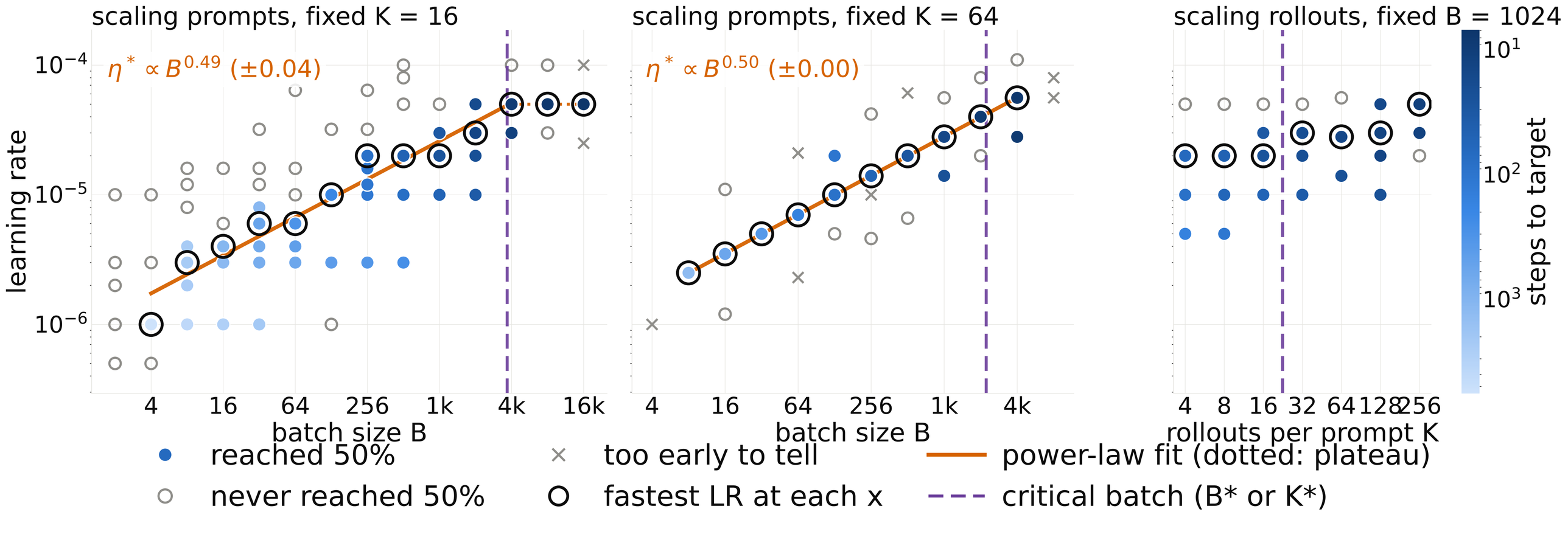
scaling prompts, fixed K = 16
$\eta^* \propto B^{0.49}$ (±0.04)
scaling prompts, fixed K = 64
$\eta^* \propto B^{0.50}$ (±0.00)
scaling rollouts, fixed B = 1024
learning rate
10^{-4}
10^{-5}
10^{-6}
4 16 64 256 1k 4k 16k
batch size B
4 16 64 256 1k 4k
batch size B
4 8 16 32 64 128 256
rollouts per prompt K
steps to target
10^1
10^2
10^3
reached 50%
never reached 50%
too early to tell
fastest LR at each x
power-law fit (dotted: plateau)
critical batch (B* or K*)